

Module 3:

Search Engines

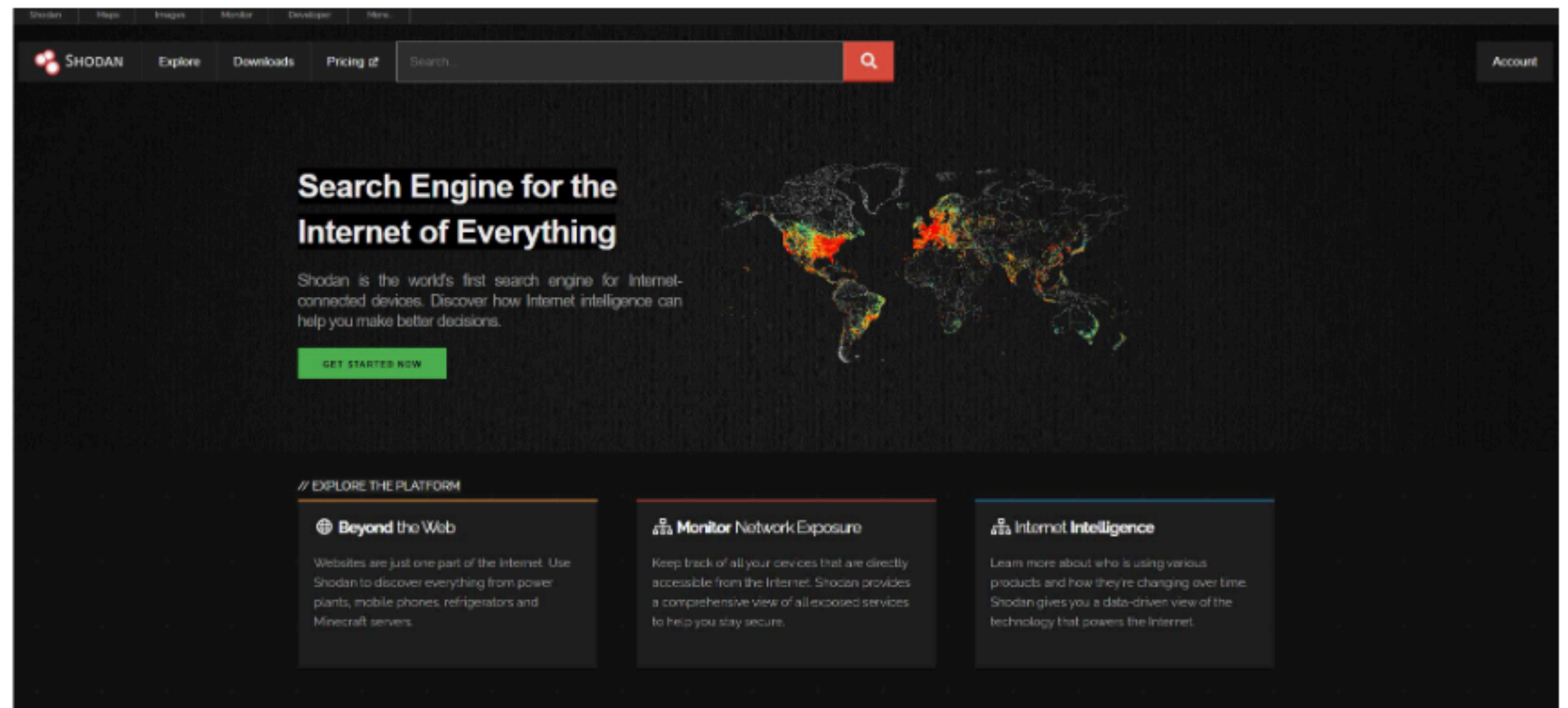


Topic 3: Shodan



What is Shodan?

<https://www.shodan.io/> is a search engine for internet-connected devices including web servers, databases, routers, printers, webcams, industrial control systems (ICS) and others.



Shodan Advantages

The information provided by shodan can be used in different areas such as :

- **Network Monitoring:** organizations can monitor their connected devices to the internet.
- **IoT Devices:** tracking the usage of IoT devices online.
- **Market Research:** business owners can track the distribution of their products on the internet and also determine what products people use the most.
- **Enterprise Security:** identifying endpoints in the enterprise system and ensuring the security of the banners.
- **Tracking Malware Spreading:** researchers can track malware spreading and how many online devices are infected.

Shodan Dorks

Shodan Dorks – (in other words: Search Filters) – are special keywords that can be used to search for specific properties and narrow down search results.

Format:

filtername:value

Notes:

- There is no space in between the filter name and its value. (filtername:value)
- If the value contains spaces, it need to be wrapped in quotes.
(filtername:"value example")
- Filters can also be combined to narrow down results even further.
(filter1:value1 filter2:value2)

Shodan Dorks

Filter	Usage
<code>country:</code>	Find devices in a specific country

Note: This filter accepts a 2-letter country code [alpha-2] (ISO 3166-1993) as an argument

Shodan Dorks

Filter	Usage
city:	Find devices in a specific city

Shodan Dorks

Filter	Usage
geo:	Find devices by giving geographical coordinates

Note: This filter accepts decimal degrees (DD) notation (latitude,longitude)

Shodan Dorks

Filter	Usage
org:	Find devices in specific organization's network

Shodan Dorks

Filter	Usage
os:	Find devices based on a specific operating system

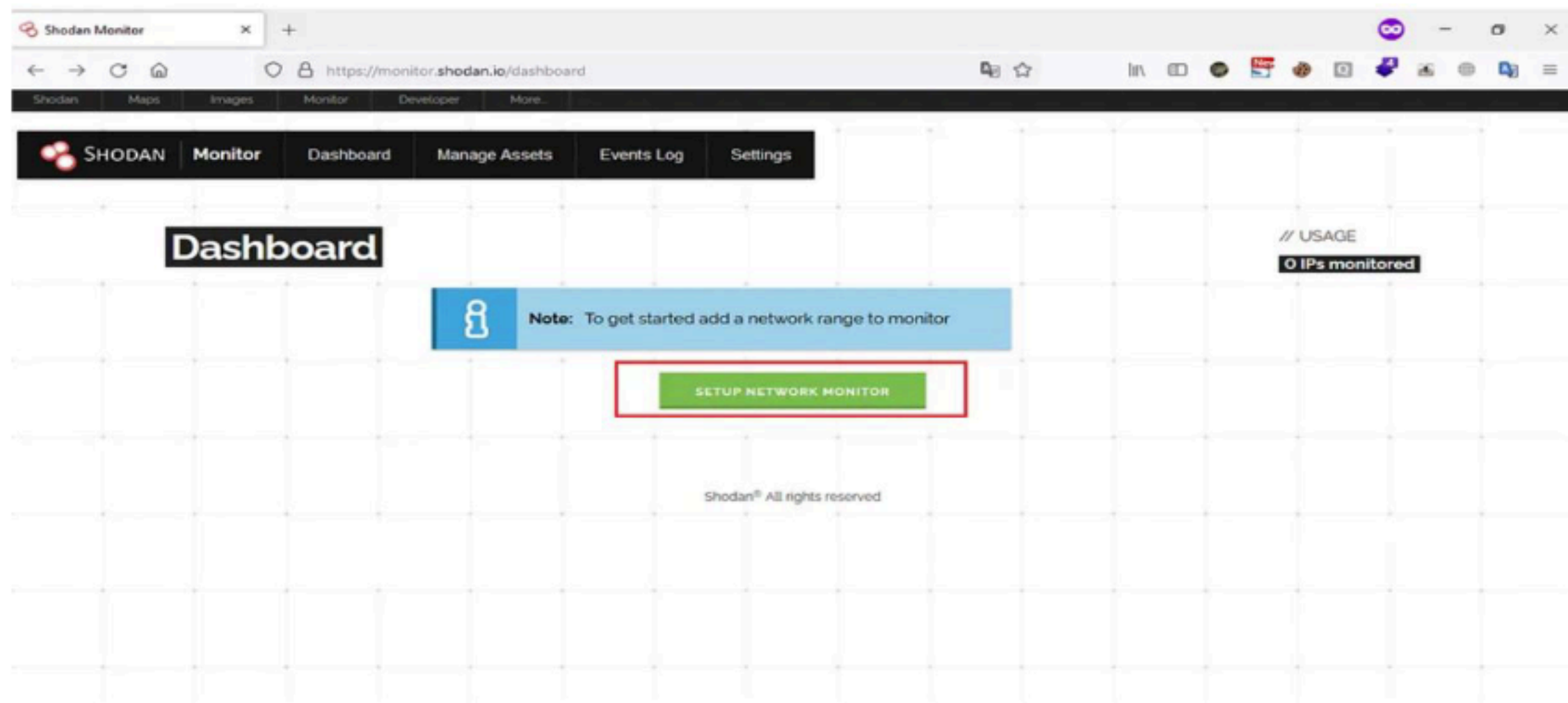
Shodan Dorks

Filter	Usage
<code>has_screenshot:</code>	Return results based on the availability of a screenshot

Note: This filter accepts (true/false) as an argument

Shodan: Create Triggers

Add the name and IP address of your target (you can also add net range like 104.16.100.52/32) and click on Add Network.



Network setup

Shodan: Create Triggers

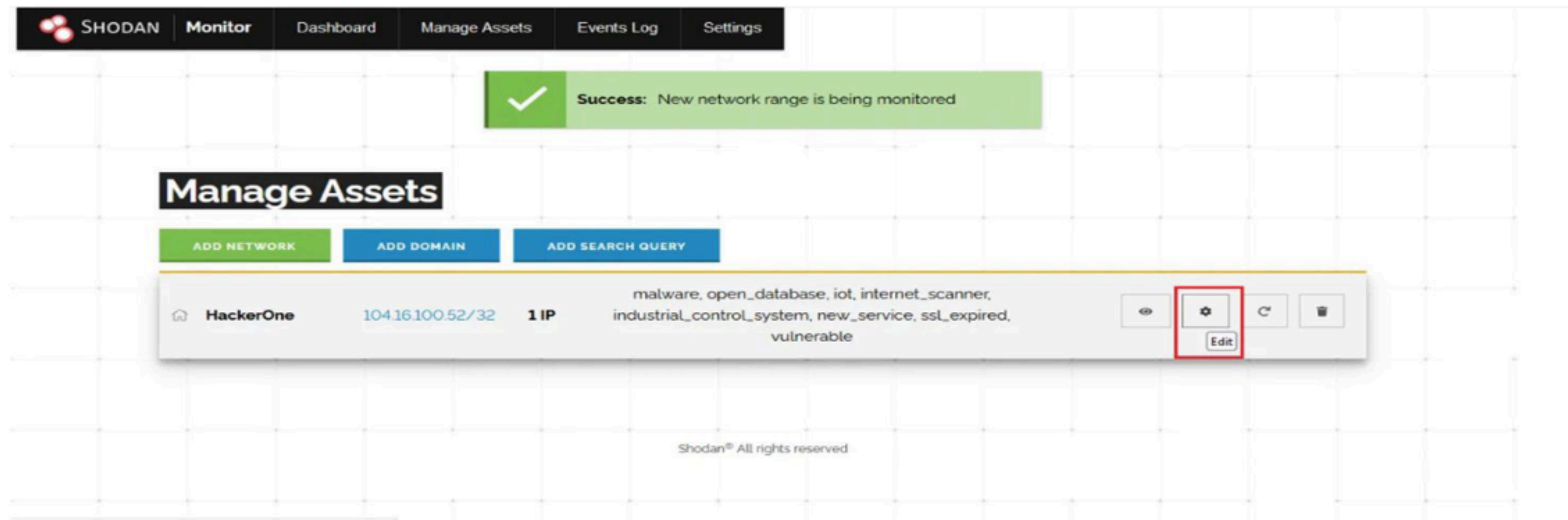
Add the name and IP address of your target (you can also add net range like 104.16.100.52/32) and click on Add Network.

The screenshot shows the 'Monitor Network' form in the Shodan Monitor web application. The form is divided into two main sections: 'General Information' and 'Notification Services'. In the 'General Information' section, there are two input fields: 'Name (ex. Production Network)' with the value 'HackerOne' and 'IP' with the value '104.16.100.52'. Below the IP field, it says 'IPs: 0'. In the 'Notification Services' section, there is a checkbox that is currently unchecked. At the bottom of the form, there is a green button labeled 'ADD NETWORK'. To the right of the form, there are two informational boxes. The top box is titled '16 IPs remaining' and states: 'Your account is allowed to monitor up to 16 IPs in total. To monitor additional IPs please upgrade your plan from the [Billing section](#).' The bottom box states: 'Select how you want to get notified. If nothing is selected then we will send an email to your account's address. To add or remove services visit [Settings](#)'.

Target Name and IP

Shodan: Create Triggers

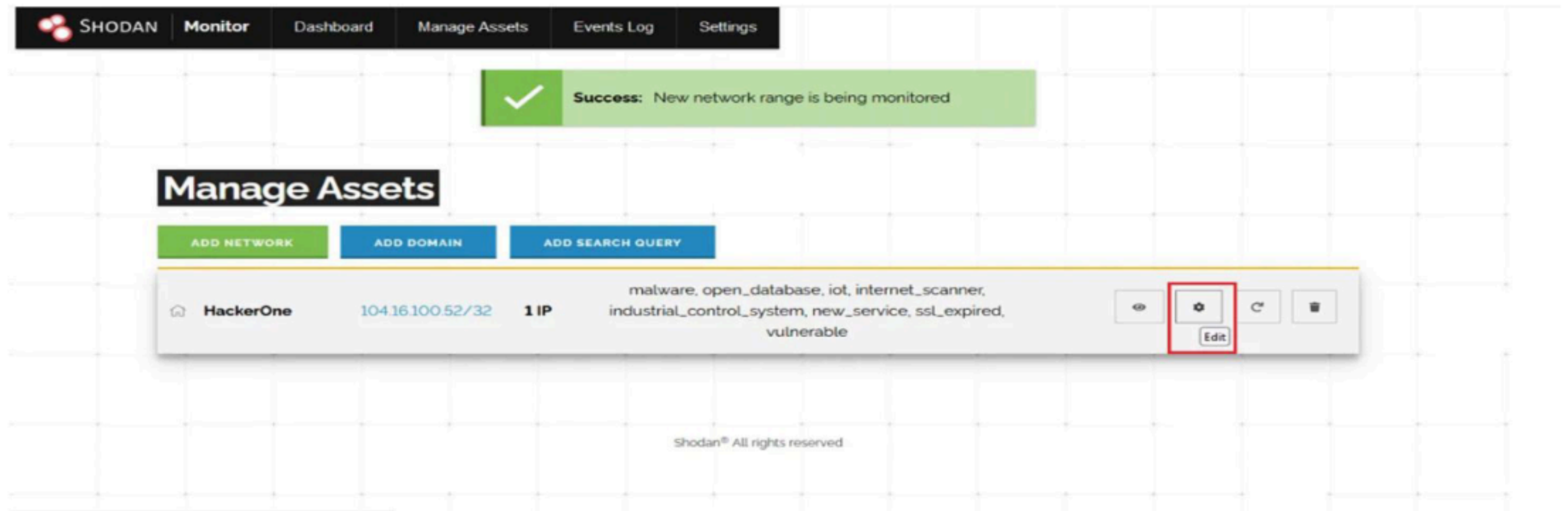
Now click on edit and scroll down to apply trigger rules and then click on Save Changes.



Editing for Trigger Rules

Shodan: Create Triggers

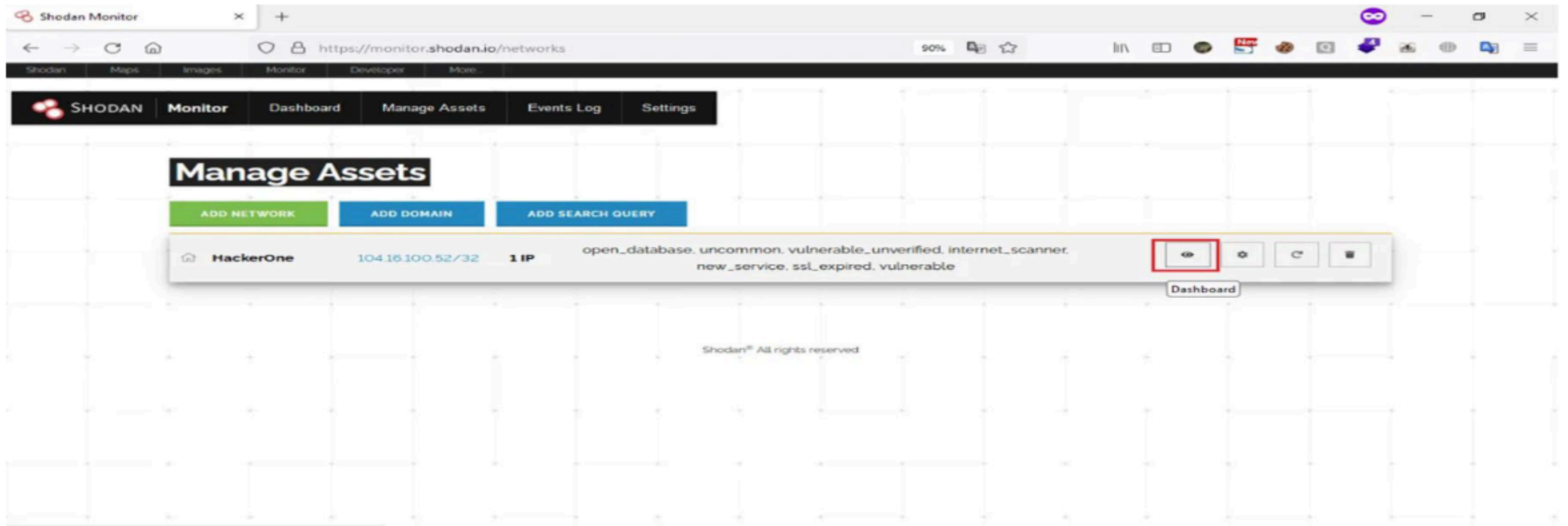
Now click on edit and scroll down to apply trigger rules and then click on Save Changes.



Editing for Trigger Rules

Shodan: Create Triggers

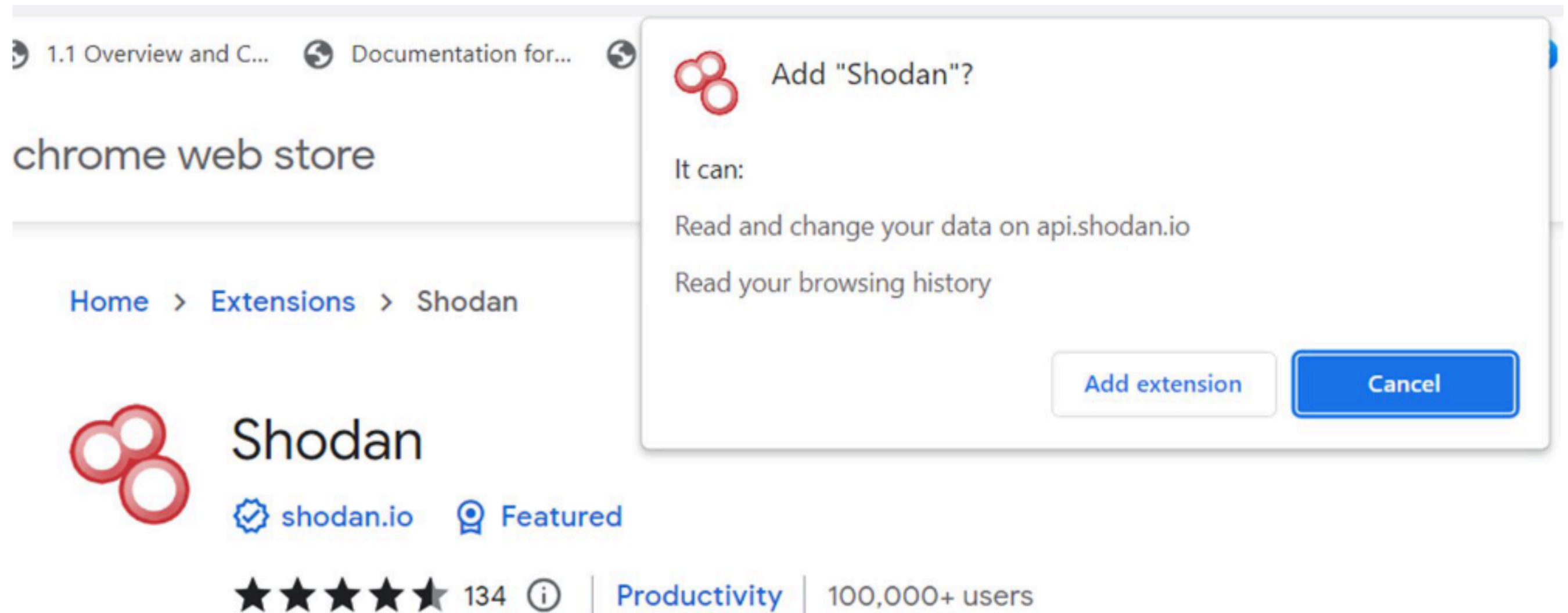
Then click on dashboard > browse results



Dashboard

Shodan Browser Plugin

Tells you where the website is hosted (country, city), who owns the IP and what other services/ ports are open.



The screenshot shows the Chrome Web Store page for the Shodan extension. The page header includes tabs for "1.1 Overview and C...", "Documentation for...", and a third tab. The breadcrumb trail reads "Home > Extensions > Shodan". The extension's name "Shodan" is displayed next to its logo, which consists of three red circles. Below the name, it says "shodan.io" with a verified badge and "Featured" with a location pin icon. The extension has a 5-star rating from 134 reviews and is categorized under "Productivity" with over 100,000 users.

1.1 Overview and C... Documentation for...

chrome web store

Home > Extensions > Shodan

 **Shodan**

 shodan.io  Featured

★★★★★ 134 ⓘ | Productivity | 100,000+ users

Add "Shodan"?

It can:

- Read and change your data on api.shodan.io
- Read your browsing history

[Add extension](#) [Cancel](#)

Shodan Browser Plugin

The screenshot shows a web browser window with the Shodan Browser Plugin overlay. The plugin displays the following information for the target **www.hackerone.com**:

- IP Address:** 104.16.100.52
- Hostname(s):** www.hackerone.com, api.hackerone.com, hackerone.com
- Country:** United States
- City:** San Francisco
- Organization:** Cloudflare, Inc.
- Open Ports:** 80, 443, 2082, 2083, 2086, 2087, 8080, 8443, 8880

Below the information, there are two buttons: **VIEW IP DETAILS** and **VIEW DOMAIN DETAILS**. At the bottom of the plugin, there is a section for "Email address" with a text input field and a **Submit** button. A cookie consent banner is visible at the bottom of the browser window.

Target Information - Shodan Browser Plugin

Shodan Dorks Hands-On

- find MongoDB database servers which have open authentication over the public internet within Shodan.

"MongoDB Server Information" port:27017 -authentication

- look for FTP servers that allow anonymous logins.

"220" "230 Login successful." port:21

109.235.189.3

- find devices running a specific version of a RouterOS operating system:

Ex.MikroTik RouterOS 6.45.9"

port:8291 os:"MikroTik RouterOS 6.45.9"

Shodan Dorks Hands-On

- open Windows Remote Desktop ports.

`remote desktop "port:3389"`

- Windows 7 devices in Egypt.

`os:"windows 7" country:"HK"`

- Telnet Access - NO password required for telnet access

`port:"23"`

Shodan Dorks Hands-On

(Note) Difference between Shodan and Google:

- The major difference between Shodan and Google is that, that Shodan analyzes the internet and Google analyzes the Word Wide Web.

“The world wide web, or web for short, are the pages you see when you're at a device and you're online. But the internet is the network of connected computers that the web works on, as well as what emails and files travel across. Think of the internet as the roads that connect towns and cities together”